

ACRBP151

## Triton X-100™

### SECTION 1. IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

**产品说明:**  
**Product Description:** 曲拉通X-100  
Triton X-100™

**Cat No. :** BP151-100; BP151-500  
**Synonyms** Polyoxyethylene(10) octylphenyl ether  
**CAS No** 9002-93-1

**Supplier** **UK entity/business name**  
Fisher Scientific UK  
Bishop Meadow Road,  
Loughborough, Leicestershire LE11 5RG, United Kingdom

**EU entity/business name**  
Thermo Fisher Scientific  
Janssen Pharmaceuticaaan 3a, 2440 Geel, Belgium

**Emergency Telephone Number** For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11  
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No. **US**:001-800-424-9300 / **Europe**:001-703-527-3887

**E-mail address** begel.sdsdesk@thermofisher.com

**Recommended Use** Laboratory chemicals.  
**Uses advised against** No Information available

### SECTION 2. HAZARD IDENTIFICATION

**Physical State**  
Liquid

**Appearance**  
Yellow

**Odor**  
Characteristic

#### Emergency Overview

Harmful if swallowed. Causes serious eye damage. Toxic to aquatic life with long lasting effects.

#### Classification of the substance or mixture

|                                   |            |
|-----------------------------------|------------|
| Acute Oral Toxicity               | Category 4 |
| Serious Eye Damage/Eye Irritation | Category 1 |
| Chronic aquatic toxicity          | Category 2 |

#### Label Elements



**Signal Word**

**Danger**

**Hazard Statements**

H302 - Harmful if swallowed

H318 - Causes serious eye damage

H411 - Toxic to aquatic life with long lasting effects

**Precautionary Statements****Prevention**

P264 - Wash face, hands and any exposed skin thoroughly after handling

P270 - Do not eat, drink or smoke when using this product

P280 - Wear protective gloves/protective clothing/eye protection/face protection

**Response**

P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing

P310 - Immediately call a POISON CENTER or doctor

P330 - Rinse mouth

**Storage**

P403 - Store in a well-ventilated place

**Disposal**

P501 - Dispose of contents/ container to an approved waste disposal plant

**Physical and Chemical Hazards**

None identified.

**Health Hazards**

Harmful if swallowed.

**Environmental hazards**

Toxic to aquatic life with long lasting effects. Will likely be mobile in the environment due to its water solubility. The product is water soluble, and may spread in water systems.

**Other Hazards**

Toxic to terrestrial vertebrates. Contains a known or suspected endocrine disruptor. Included in the list established in accordance with Article 59(1) for having endocrine disrupting properties. Contains a substance on the National Authorities Endocrine Disruptor Lists.

**SECTION 3. COMPOSITION/INFORMATION ON INGREDIENTS**

| Component                                                                               | CAS No    | Weight % |
|-----------------------------------------------------------------------------------------|-----------|----------|
| Poly(oxy-1,2-ethanediyl), .alpha.-[4-(1,1,3,3-tetramethylbutyl)phenyl]-.omega.-hydroxy- | 9002-93-1 | <=100    |

**SECTION 4. FIRST AID MEASURES****General Advice**

If symptoms persist, call a physician.

**Eye Contact**

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.

**Skin Contact**

Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician.

**Inhalation**

Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.

**Ingestion**

Clean mouth with water and drink afterwards plenty of water.

**Most important symptoms and effects**

Causes severe eye damage. Causes eye burns.

**Self-Protection of the First Aider**

Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.

**Notes to Physician**

Treat symptomatically.

**SECTION 5. FIRE-FIGHTING MEASURES****Suitable Extinguishing Media**

Water spray, carbon dioxide (CO<sub>2</sub>), dry chemical, alcohol-resistant foam.

**Extinguishing media which must not be used for safety reasons**

Do not use a solid water stream as it may scatter and spread fire.

**Specific Hazards Arising from the Chemical**

Thermal decomposition can lead to release of irritating gases and vapors. Keep product and empty container away from heat and sources of ignition.

**Protective Equipment and Precautions for Firefighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

**SECTION 6. ACCIDENTAL RELEASE MEASURES****Personal Precautions**

Use personal protective equipment as required. Ensure adequate ventilation.

**Environmental Precautions**

Do not flush into surface water or sanitary sewer system.

**Methods for Containment and Clean Up**

Soak up with inert absorbent material. Keep in suitable, closed containers for disposal.

Refer to protective measures listed in Sections 8 and 13.

**SECTION 7. HANDLING AND STORAGE****Handling**

Wear personal protective equipment/face protection. Ensure adequate ventilation. Do not get in eyes, on skin, or on clothing. Avoid ingestion and inhalation.

**Storage**

Keep containers tightly closed in a dry, cool and well-ventilated place.

**Specific Use(s)**

Use in laboratories

**SECTION 8. EXPOSURE CONTROLS/PERSONAL PROTECTION****Control Parameters****Exposure Controls****Engineering Measures**

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Ensure adequate ventilation, especially in confined areas. Ensure that eyewash stations and safety showers are close to the workstation location. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source.

## Personal protective equipment

**Eye Protection** Wear safety glasses with side shields (or goggles) Goggles (European standard - EN 166)

**Hand Protection** Protective gloves

| Glove material | Breakthrough time | Glove thickness | EU standard | Glove comments        |
|----------------|-------------------|-----------------|-------------|-----------------------|
| Natural rubber | See manufacturers | -               | EN 374      | (minimum requirement) |
| Nitrile rubber | recommendations   |                 |             |                       |
| Neoprene       |                   |                 |             |                       |
| PVC            |                   |                 |             |                       |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatibility, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

**Skin and body protection** Long sleeved clothing

**Respiratory Protection** No special protective equipment required.

**Large scale/emergency use** In case of insufficient ventilation, wear suitable respiratory equipment  
**Recommended Filter type:** Particulates filter conforming to EN 143

**Small scale/Laboratory use** No personal respiratory protective equipment normally required. Maintain adequate ventilation.

**Hygiene Measures** Handle in accordance with good industrial hygiene and safety practice.

**Environmental exposure controls** Prevent product from entering drains. Do not allow material to contaminate ground water system.

## SECTION 9. PHYSICAL AND CHEMICAL PROPERTIES

|                                   |                   |                                 |
|-----------------------------------|-------------------|---------------------------------|
| <b>Appearance</b>                 | Yellow            |                                 |
| <b>Physical State</b>             | Liquid            |                                 |
| <b>Odor</b>                       | Characteristic    |                                 |
| <b>Odor Threshold</b>             | No data available |                                 |
| <b>pH</b>                         | 6-8               | 5% aq.sol                       |
| <b>Melting Point/Range</b>        | 6 °C / 42.8 °F    |                                 |
| <b>Softening Point</b>            | No data available |                                 |
| <b>Boiling Point/Range</b>        | 270 °C / 518 °F   | @ 760 mmHg                      |
| <b>Flash Point</b>                | 251 °C / 483.8 °F | <b>Method -</b> CC (closed cup) |
| <b>Evaporation Rate</b>           | No data available |                                 |
| <b>Flammability (solid,gas)</b>   | Not applicable    | Liquid                          |
| <b>Explosion Limits</b>           | No data available |                                 |
| <b>Vapor Pressure</b>             | No data available |                                 |
| <b>Vapor Density</b>              | No data available | (Air = 1.0)                     |
| <b>Specific Gravity / Density</b> | 1.067             |                                 |
| <b>Bulk Density</b>               | Not applicable    | Liquid                          |
| <b>Water Solubility</b>           | Soluble           |                                 |

|                                                                                                |                          |
|------------------------------------------------------------------------------------------------|--------------------------|
| <b>Solubility in other solvents</b>                                                            | No information available |
| <b>Partition Coefficient (n-octanol/water)</b>                                                 |                          |
| <b>Component</b>                                                                               | <b>log Pow</b>           |
| Poly(oxy-1,2-ethanediyl),<br>.alpha.-[4-(1,1,3,3-tetramethylbutyl)phe<br>nyl]-.omega.-hydroxy- | 2.7                      |
| <b>Autoignition Temperature</b>                                                                | No data available        |
| <b>Decomposition Temperature</b>                                                               | No data available        |
| <b>Viscosity</b>                                                                               | No data available        |
| <b>Explosive Properties</b>                                                                    | No information available |
| <b>Oxidizing Properties</b>                                                                    | No information available |

**SECTION 10. STABILITY AND REACTIVITY**

|                                         |                                                                                                       |
|-----------------------------------------|-------------------------------------------------------------------------------------------------------|
| <b>Stability</b>                        | Stable under normal conditions.                                                                       |
| <b>Hazardous Reactions</b>              | None under normal processing.                                                                         |
| <b>Hazardous Polymerization</b>         | Hazardous polymerization does not occur.                                                              |
| <b>Conditions to Avoid</b>              | Incompatible products. Excess heat. Exposure to air. Exposure to light. Exposure to moisture.         |
| <b>Materials to avoid</b>               | Strong oxidizing agents. Strong acids. Strong reducing agents.                                        |
| <b>Hazardous Decomposition Products</b> | Carbon monoxide (CO). Carbon dioxide (CO <sub>2</sub> ). Aldehydes. Ketones. Formaldehyde. peroxides. |

**SECTION 11. TOXICOLOGICAL INFORMATION****Product Information****(a) acute toxicity;**

| Component                                                                                      | LD50 Oral          | LD50 Dermal | LC50 Inhalation |
|------------------------------------------------------------------------------------------------|--------------------|-------------|-----------------|
| Poly(oxy-1,2-ethanediyl),<br>.alpha.-[4-(1,1,3,3-tetramethylbutyl)phenyl]-<br>.omega.-hydroxy- | 1800 mg/kg ( Rat ) |             |                 |

**(b) skin corrosion/irritation;** No data available

**(c) serious eye damage/irritation;** Category 1

**(d) respiratory or skin sensitization;**

|                    |                   |
|--------------------|-------------------|
| <b>Respiratory</b> | No data available |
| <b>Skin</b>        | No data available |

**(e) germ cell mutagenicity;** No data available

**(f) carcinogenicity;** No data available

There are no known carcinogenic chemicals in this product

**(g) reproductive toxicity;** No data available

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|                                            |                          |
|--------------------------------------------|--------------------------|
| (h) STOT-single exposure;                  | No data available        |
| (i) STOT-repeated exposure;                | No data available        |
| Target Organs                              | None known.              |
| (j) aspiration hazard;                     | No data available        |
| Symptoms / effects, both acute and delayed | No information available |

## SECTION 12. ECOLOGICAL INFORMATION

**Ecotoxicity effects** The product contains following substances which are hazardous for the environment. Toxic to aquatic organisms, may cause long-term adverse effects in the aquatic environment.

| Component                                                                                      | Freshwater Fish                                                     | Water Flea         | Freshwater Algae | Microtox |
|------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|--------------------|------------------|----------|
| Poly(oxy-1,2-ethanediyl),<br>.alpha.-[4-(1,1,3,3-tetramethylbutyl)phenyl]-<br>.omega.-hydroxy- | LC50 = 8.9 mg/L 96H<br>LC50 = 4.0 mg/l 96H<br>(Pimephales promelus) | EC50 = 26 mg/L 48h | -                | -        |

**Persistence and Degradability**

**Persistence** Persistence is unlikely.

| Component                                                                                                         | Degradability |
|-------------------------------------------------------------------------------------------------------------------|---------------|
| Poly(oxy-1,2-ethanediyl),<br>.alpha.-[4-(1,1,3,3-tetramethylbutyl)phenyl]-.omega.-hydroxy-<br>9002-93-1 ( <=100 ) | 60% >28 days  |

**Degradation in sewage treatment plant** Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants.

**Bioaccumulative Potential** Bioaccumulation is unlikely

| Component                                                                                      | log Pow | Bioconcentration factor (BCF) |
|------------------------------------------------------------------------------------------------|---------|-------------------------------|
| Poly(oxy-1,2-ethanediyl),<br>.alpha.-[4-(1,1,3,3-tetramethylbutyl)phenyl]-<br>.omega.-hydroxy- | 2.7     | No data available             |

**Mobility in soil** The product is water soluble, and may spread in water systems Will likely be mobile in the environment due to its water solubility Highly mobile in soils

**Endocrine Disruptor Information**

| Component                                                                                      | EU - Endocrine Disruptors Candidate List | EU - Endocrine Disruptors - Evaluated Substances | Japan - Endocrine Disruptor Information |
|------------------------------------------------------------------------------------------------|------------------------------------------|--------------------------------------------------|-----------------------------------------|
| Poly(oxy-1,2-ethanediyl),<br>.alpha.-[4-(1,1,3,3-tetramethylbutyl)phenyl]-<br>.omega.-hydroxy- | Group III Chemical                       | -                                                | -                                       |

**Persistent Organic Pollutant** This product does not contain any known or suspected substance  
**Ozone Depletion Potential** This product does not contain any known or suspected substance

## SECTION 13. DISPOSAL CONSIDERATIONS

**Waste from Residues/Unused Products** Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations. Should not be released into the environment.

**Contaminated Packaging** Dispose of this container to hazardous or special waste collection point.

**Other Information**

Do not flush to sewer. Waste codes should be assigned by the user based on the application for which the product was used. Do not empty into drains. Do not let this chemical enter the environment.

**SECTION 14. TRANSPORT INFORMATION****Road and Rail Transport**

**UN-No** UN3082  
**Proper Shipping Name** ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S.  
**Technical Shipping Name** Octylphenyl-polyethylene glycol  
**Hazard Class** 9  
**Packing Group** III

**IMDG/IMO**

**UN-No** UN3082  
**Proper Shipping Name** ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S.  
**Technical Shipping Name** Octylphenyl-polyethylene glycol  
**Hazard Class** 9  
**Packing Group** III

**IATA**

**UN-No** UN3082  
**Proper Shipping Name** ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S.  
**Technical Shipping Name** Octylphenyl-polyethylene glycol  
**Hazard Class** 9  
**Packing Group** III

**Special Precautions for User** No special precautions required

**SECTION 15. REGULATORY INFORMATION****International Inventories**

X = listed.

| Component                                                                                        | The Inventory of Hazardous Chemicals (2015 Edition) | List of dangerous goods GB 12268 - 2012 | TCSI | IECSC | EINECS | TSCA | DSL | PICCS | ENCS | ISHL | AICS | KECL     |
|--------------------------------------------------------------------------------------------------|-----------------------------------------------------|-----------------------------------------|------|-------|--------|------|-----|-------|------|------|------|----------|
| Poly(oxy-1,2-ethanediyl),<br>alpha-[4-(1,1,3,3-tetra<br>methylbutyl)phenyl]-.o<br>mega.-hydroxy- | -                                                   | -                                       | X    | X     | -      | X    | X   | X     | X    | X    | X    | KE-33568 |

**National Regulations****SECTION 16. OTHER INFORMATION**

**Creation Date** 22-Nov-2010

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**Revision Date** 20-Apr-2024  
**Revision Summary** SDS sections updated.

**Training Advice**

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

**Legend**

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**DNEL** - Derived No Effect Level

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**NOEC** - No Observed Effect Concentration

**PBT** - Persistent, Bioaccumulative, Toxic

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

**PNEC** - Predicted No Effect Concentration

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**POW** - Partition coefficient Octanol:Water

**vPvB** - very Persistent, very Bioaccumulative

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (Volatile Organic Compound)

**Key literature references and sources for data**

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**